1.2.1 Network Specification

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Title page

Danish Business Authority

Nemhandel eDelivery Network Specification version 1.2.0

Introduction

Nemhandel eDelivery is a public network for transferring business documents. It is an implementation of the eDelivery AS4 - 1.15 specification. Nemhandel eDelivery is a peer-to-peer network with centralized discovery, namely, the SML and the SMP. AS4 is used as the transport protocol between access points. Nemhandel eDelivery is a national network that enables companies to fulfil their obligations according to Danish law (Bogføringsloven).

Nemhandel eDelivery is closely modeled on the Peppol eDelivery Network specifications, and follows the Peppol specifications with some exceptions and additions. The following Peppol specifications are the foundation for Nemhandel eDelivery:

Peppol SML version 1.2.0
Peppol SMP version 1.3.0
Peppol Policy for use of identifiers version 4.2.0
Peppol SBDH Envelope specification version 2.0.1
Peppol AS4 profile version 2.0.2
Peppol Policy for Transport Security version 1.1.0

The following Peppol specifications are not used by Nemhandel eDelivery:

Peppol AS2 profile
Peppol Directory

Error codes

Access points MUST use Nemhandel eDelivery error code list version 1.2.0 when reporting errors to other access points, and only these.

The specific error codes mentioned in this specification MUST be used when reporting the associated errors.

Document formats

Documents that are transferred in the network MUST be XML SBDH documents that can be validated according to section: Validation.

Payload

The payload of a SBDH document is defined in this specification to be the single XML element that is a child of the <StandardBusinessDocument> element and not the <StandardBusinessDocumentHeader> element nor any of its descendents.

Identification schemes

Nemhandel eDelivery requires that the payload of every document sent in the network must be signed by a party to the transaction. The party that has signed a given document must be identifiable. This identification is based on public key certificates.

An access point MUST use a supported certificate for signing business documents. Currently only MitID system certificates are supported (FOCES3 also known as "VOCES3 - FOCES profile").

MitID system certificates

MitID system certificates (FOCES3) are issued by the Danish State (issuer=/O=Den Danske Stat).

A MitID system certificate used in the network for signing documents MUST identify a Danish company by its CVR number.

CVR numbers are recorded in the subject string of the certificate. Example for CVR number 10150817:

subject=/2.5.4.97=NTRDK-10150817

Document signatures

All documents transferred in the network MUST contain a digital signature of the payload.

The signing party MAY be the original creator of the payload.

The signing party MAY be an intermediary (C1 in the four-corner model).

The signing party MAY be the sending access point (C2 in the four-corner model).

The signing party MUST NOT be the receiving access point (C3 in the four-corner model).

The signing party MUST NOT be any other party (e.g. C4 in the four-corner model)

All signatures MUST use a certificate conforming to section: Identification schemes.

A document signature MUST be in XAdES format compliant with ETSI EN 319 132-1 V1.2.1 (XAdES). The document signature MUST be created and validated using signature class/level AdES-BASELINE-B (ETSI EN 319 102-1 V1.3.1). As signature method, http://www.w3.org/2001/04/xmldsig-more#rsa-sha256 MUST be used, and as digest method, http://www.w3.org/2001/04/xmlenc#sha256 MUST be used. The document signature MUST contain a digest of the payload (detached signature). The document signature MUST NOT contain a copy of the entire payload.

The the digest of the payload MUST be generated from the canonicalized form of the payload. The canonicalization method MUST be http://www.w3.org/2001/10/xml-exc-c14n.

The XAdES encoded document signature MUST be placed in the Standard Business Document Header under the <BusinessScope> <Scope> element. The <Type> of the scope MUST have the value "NEMHANDEL_EDELIVERY_SIGNATURE". The XAdES encoded document signature MUST be Base64 encoded and placed as the value of the <InstanceIdentifier> element. An additional element <Identifier> with the value "nemhandel-edelivery-1.2" MUST be placed alongside (E-APR21023).

Example:

SBD Example

```
<StandardBusinessDocument xmlns="http://www.unece.org/cefact/namespaces/StandardBusinessDocumentHeader">
   <StandardBusinessDocumentHeader>
      ...
      <BusinessScope>
         ...
         <Scope>
            <Type>NEMHANDEL_EDELIVERY_SIGNATURE</Type>
            <InstanceIdentifier> <!-- Base64 encoded XAdES signature document --> </InstanceIdentifier>
            <Identifier>nemhandel-edelivery-1.2</Identifier>
         </Scope>
      </BusinessScope>
   </StandardBusinessDocumentHeader>
   ... <!-- the payload is an element placed here -->
</StandardBusinessDocument>
```

Header content

There MUST only be one NEMHANDEL_EDELIVERY_SIGNATURE scope in a given SBDH.

All documents transfered in the network MUST be identified with a unique identifier in the SBDH.

The unique identifer SHOULD be in UUID version 4 format.

The unique identifer MUST be placed as the value of the element <StandardBusinessDocument> <StandardBusinessDocumentHeader> <DocumentIdentification> <InstanceIdentifier> in the SBDH.

Sender and reciever identifiers

The sender identifier MUST be placed as the value of the element <StandardBusinessDocument> <StandardBusinessDocumentHeader> <Sender><Identifier> in the SBDH.

The reciever identifier MUST be placed as the value of the element <StandardBusinessDocument> <StandardBusinessDocumentHeader> <Reciever><Identifier> in the SBDH.

Sender and receiver identifiers MUST include an indentifier scheme id and value.

CVR numbers MUST include 8-digits without any DK prefix in the identifier value.

CVR number example

```
...
<Sender>
                        <Identifier Authority="iso6523-actorid-upis">0184:16356706</Identifier>
</Sender>
...
```

GLN number example

```
...
<Receiver>
                        <Identifier Authority="iso6523-actorid-upis">0088:5798009811639</Identifier>
</Receiver>
...
```

Validation

Content validation

Upon receiving a message the payload MUST be validated according to its XSD.

If a message payload is not valid according to the specified XSD for the given document type, it MUST be rejected immediately (synchronously) according to the AS4 protocol (E-APR21016).

If a message payload is valid according to the specified XSD for the given document type it MUST be accepted according to the AS4 protocol.

If a message is accepted, its payload content MUST be validated according to specific business rules, e.g. Schematron rules.

If the message payload content is not valid according to Schematron rules or equivalent, the message MUST be rejected, and the error MUST be specified in an asynchronous response to the sender (E-APR21014).

Asynchronous responses

If the message payload is in Peppol BIS format, an asynchronous response MUST be a Peppol BIS Message Level Response (MLR).

If the message payload is in OIOUBL version 3.x format, an asynchronous response MUST be a OIOUBL Message Level Response (MLR).

If the message payload is in OIOUBL version 2.x format, an asynchronous response MUST be an OIOUBL Application Response (AR).

An asynchronous response to a message MUST contain a reference to the message's unique identifier in the path DocumentResponse /DocumentReference/ID.

A message that is an asynchronous response MUST NOT be replied to with another asynchronous response.

A receiver MAY send an asynchronous response as an acknowledgement of reception of a valid message.

Signature validation

An access point receiving a message from another access point MUST verify that it contains a valid document signature in the header.

A message received from a party other than an access point MAY contain a valid document signature in the header.

If a message received from another access point does not contain a signature, or contains more than one signature, the access point MUST reject the message (E-APR21020, APR21022).

If a message contains an invalid signature, the access point MUST reject the message (E-APR21021).

If an access point rejects a message from another access point due to signature validation error, it MUST do so with an asynchronous response to the sender.

An XAdES document signature MUST conform to the structure for creating document signatures specified in section: Document signatures (E-APR21021).

Given a message that contains a document signature, the message MUST match the document signature according to this rule: The calculated hash value of the canonicalized form of the payload must be identical to the XAdES payload hash value in the document signature. The canonicalization method used is Exclusive XML Canonicalization (http://www.w3.org/2001/10/xml-exc-c14n). (E-APR21021).

The certificate used for document signing MUST be verified to be valid. If a certificate is part of a PKI that has a Certificate Revocation List (CRL), OCSP server or equivalent, at least one of these MUST be checked as part of validation (E-APR21021).

SML

Nemhandel eDelivery uses SML DK for address discovery: https://sml.erst.dk/nemhandel-sml/

SMP

All senders MUST be registered with an access point in an eDelivery SMP (Nemhandelregistret or third party SMP) with the transport profile "nemhandel-transport-as4-v2_0", process id "oioubl-procid-ubl::Procurement-BilSim-1.0" and document type id "busdox-docid-qns::urn:oasis:names:specification:ubl:schema:xsd:ApplicationResponse-2::ApplicationResponse##OIOUBL-2.1::2.1". This ensures that the sender can receive asynchronous responses.

All senders SHOULD look up the sender identification of a SBDH in the SML/SMP before sending to ensure the sender can receive asynchronous responses. The message SHOULD be rejected by the sender if the sender identification registration is incorrect.The SMP lookup MAY be cached.

Access Points

All access points MUST comply with Nemhandel eDelivery Access Point Specification version 1.2.0